

Closed-Form Flow Matching: High-Dimensional Statistics and the Softmax Collapse

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Flow matching has emerged as a powerful framework for training continuous normalizing flows. In its Gaussian-OT instantiation, the **marginal velocity field** admits a closed form:

it is a softmax-weighted average of per-example conditional velocities. In high dimension, a striking phenomenon occurs: the softmax weights become almost one-hot very early along the interpolation path. We refer to this as **softmax collapse**. It implies that the learned flow rapidly behaves like a nearest-neighbor map and tends to memorize the training set instead of sampling from the underlying data distribution.

In this post, building on the closed-form analysis of Bertrand et al. (2025), we study this collapse phenomenon through the lens of high-dimensional statistics and statistical physics. We view the softmax weights as a Gibbs measure over the training examples, with an energy that depends on the interpolation time t . This connects the behavior of flow matching to classic models such as the Random Energy Model (REM) and to Large Deviation Principles (LDP).

Our main goal is twofold:

1. **Mechanism.** Explain why, in high dimension, the marginal velocity field very quickly aligns with the conditional velocity field even for small t , and why neural networks find certain time regimes (around $t \approx 0.1$ and near $t \approx 1$) particularly hard to learn.
2. **Prediction.** Derive a **predictive formula for the collapse time**—the value of t at which the softmax weights become essentially one-hot in a CIFAR-10 proxy.

We adapt the dynamical and statistical-physics framework of Biroli et al. (2024) to the flow-matching setting. Treating the softmax energies as REM-like random energies, we obtain an explicit asymptotic expression for the free energy, which in turn controls when the planted datapoint wins over all competing training samples. Specializing this analysis to a Gaussian proxy calibrated to CIFAR-10 yields a concrete collapse-time prediction.

Contributions

- We interpret the closed-form marginal velocity field in flow matching as a Gibbs measure
- Within an i.i.d. Gaussian toy model calibrated to CIFAR-10, we derive an **explicit high-dimensional formula for the collapse time** as a function of the dataset dimension, sample size, and variance, and give its numerical specialization for CIFAR-10.
- We provide a **self-contained, pedagogical derivation** of the relevant LDP and REM results, aiming to be more accessible than Bertrand et al. (2025) and to serve as a reference for readers interested in using these tools in generative modeling.

In the section Section we briefly recall the flow-matching and conditional flow-matching formulations and fix notation. In Section , we summarize the empirical behavior reported in Bertrand et al. (2025). The section Section justifies the Gaussian toy model used in the theory. The core statistical-physics analysis appears in Section , and Section connects the collapse picture back to the observed error profile of learned velocity fields. We include an accompanying notebook that numerically analyzes the CIFAR-10 dataset, computing both the collapse times and the correlation between the marginal and conditional velocity fields.

Setup and Notation

We follow the notation and setup of (Lipman et al. 2024, sec. 2) to present Flow Matching (FM). Given a training dataset with samples from an unknown target distribution q over $\mathbb{R}^d$, the goal is to learn a model that can generate new samples from q . Flow Matching (FM) does this by defining a **continuous probability path** $(p_t)_{0 \leq t \leq 1}$ that morphs a **known source distribution** $p_0 = p$ (typically noise) into the **data distribution** $p_1 = q$. FM learns a **time-dependent velocity field** u_t (a vector field) that describes how samples move along this path. After training, sampling from q works by:

1. Draw $X_0 \sim p$ (easy noise).
2. Solve an **ODE** driven by the learned velocity field to transport X_0 to X_1 , which should follow q .

FM uses a time-dependent vector field: $u : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ and defines the associated **flow map** ψ_t by the ODE:

$$\frac{d}{dt}\psi_t(x) = u_t(\psi_t(x)), \quad \psi_0(x) = x.$$

If u_t generates the probability path p_t , then pushing forward the source sample $X_0 \sim p_0$ through the flow gives: $X_t := \psi_t(X_0) \sim p_t$. In particular, integrating to $t = 1$ yields: $X_1 = \psi_1(X_0) \approx q$. So the core objective is: **learn a neural velocity field** u_t^θ whose flow maps p_0 to p_1 . Ideally, FM would minimize the mean-squared error between the learned and true velocity fields:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, X_t} \left[\|u_t^\theta(X_t) - u_t(X_t)\|^2 \right], \quad t \sim \mathcal{U}[0, 1], \quad X_t \sim p_t.$$

But the marginal velocity field u_t is usually hard to compute. Let the source be standard Gaussian: $p := p_0 = \mathcal{N}(0, I)$. A convenient path can be built by mixing **conditional paths** $p_{t|1}(x|x_1)$, each conditioned on a data example $X_1 = x_1$. The resulting marginal path is:

$$p_t(x) = \int p_{t|1}(x|x_1)q(x_1)dx_1, \quad \text{where} \quad p_{t|1}(x|x_1) = \mathcal{N}(x \mid tx_1, (1-t)^2I).$$

A simple way to sample from this p_t is:

$$X_t = tX_1 + (1-t)X_0 \sim p_t,$$

where $X_0 \sim p$ and $X_1 \sim q$. Condition on a randomly chosen training sample $X_1 = x_1$ and define:

$$X_{t|1} = tx_1 + (1-t)X_0 \sim p_{t|1}(\cdot|x_1).$$

For this conditional Gaussian path, the ODE gives a **simple closed-form conditional velocity field**:

$$u_t(x|x_1) = \frac{x_1 - x}{1-t}.$$

This yields a tractable loss:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, X_t, X_1} \left[\|u_t^\theta(X_t) - u_t(X_t | X_1)\|^2 \right], \quad t \sim \mathcal{U}[0, 1], \quad X_0 \sim p, \quad X_1 \sim q,$$

with $X_t = (1 - t)X_0 + tX_1$. Crucially, this conditional objective provides the **same gradients** as the original FM loss:

$$\nabla_\theta \mathcal{L}_{\text{FM}}(\theta) = \nabla_\theta \mathcal{L}_{\text{CFM}}(\theta).$$

Plugging the conditional velocity into the loss gives a common practical form:

$$\mathcal{L}_{\text{CFM}}^{\text{OT, Gauss}}(\theta) = \mathbb{E}_{t, X_0, X_1} \left[\|u_t^\theta(X_t) - (X_1 - X_0)\|^2 \right], \quad t \sim \mathcal{U}[0, 1], \quad X_0 \sim \mathcal{N}(0, I), \quad X_1 \sim q,$$

where $X_t = (1 - t)X_0 + tX_1$.

The CFM and FM losses are related by a bias-variance decomposition. The CFM loss can be written as:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathcal{L}_{\text{FM}}(\theta) + \mathbb{E}_{t, X_t} [\text{Var}(X_1 - X_0 | X_t)].$$

i Proof of the loss decomposition

At the interpolant $X_t = (1 - t)X_0 + tX_1$, the conditional velocity field evaluates to:

$$u_t(X_t | X_1) = \frac{X_1 - X_t}{1 - t} = X_1 - X_0.$$

The marginal velocity field at X_t is the conditional expectation of this:

$$u_t(X_t) = \mathbb{E}[X_1 - X_0 | X_t].$$

Let $Y = X_1 - X_0$ denote the target and $\bar{Y} = \mathbb{E}[Y | X_t] = u_t(X_t)$ the optimal predictor. We decompose:

$$\|u_t^\theta(X_t) - Y\|^2 = \|(u_t^\theta(X_t) - \bar{Y}) + (\bar{Y} - Y)\|^2.$$

Expanding and taking the conditional expectation, the cross-term vanishes and we obtain the desired decomposition.

The second term is the **irreducible variance** — it does not depend on θ and represents the inherent uncertainty in predicting the conditional velocity $X_1 - X_0$ from X_t alone. This variance captures how much the target $X_1 - X_0$ fluctuates across different (X_0, X_1) pairs that pass through the same interpolant X_t .

Observations on the Marginal Velocity Field, Conditional Velocity Field and Training Difficulties

In this section, we present some observations made in Bertrand et al. (2025). We focus on the case where the source distribution p_0 is the standard normal distribution and the data distribution p_{data} is the CIFAR-10 dataset. We observe n datapoints from CIFAR-10:

$$x^{(1)}, \dots, x^{(n)} \sim p_{\text{data}}, \quad x^{(i)} \in \mathbb{R}^d.$$

In that scenario, we do not have access to the data distribution p_{data} , but only to the empirical distribution:

$$\hat{p}_{\text{data}} := \frac{1}{n} \sum_{i=1}^n \delta_{x^{(i)}}. \quad (1)$$

The marginal velocity field becomes available in closed form:

$$u_t(x) = \sum_{i=1}^n \lambda_i(x, t) \frac{x^{(i)} - x}{1 - t}, \quad \lambda_i(x, t) = \text{softmax} \left(-\frac{\|x - tx^{(i)}\|^2}{2(1 - t)^2} \right).$$

where the softmax is taken over $i \in \{1, \dots, n\}$, i.e.

$$\lambda_i(x, t) = \frac{\exp \left(-\frac{\|x - tx^{(i)}\|^2}{2(1 - t)^2} \right)}{\sum_{j=1}^n \exp \left(-\frac{\|x - tx^{(j)}\|^2}{2(1 - t)^2} \right)}.$$

Bertrand et al. (2025) focuses on the memorization paradox, the fact that in that context with the empirical distribution Equation 1, if the learning procedure outputs the marginal vector field, we recovers exactly the n datapoints and not random samples from p_{data} .

In their study, they study the relation between the marginal velocity field available in closed form, the conditional velocity field and the empirical velocity field learned by a neural network. We focus on this aspect of their work.

They have 2 main observations:

1. As the dimension grows, the marginal velocity field is rapidly correlated (even when t is very small) to the conditional velocity field.
2. The neural network has difficulties to learn the marginal velocity field for t close to 0.1 and close to 1.

We will expand on these observations in the next sections. The objective of this blog post is to provide some orders of magnitude of the softmax weights of the marginal velocity field, in particular in high dimension for CIFAR-10.

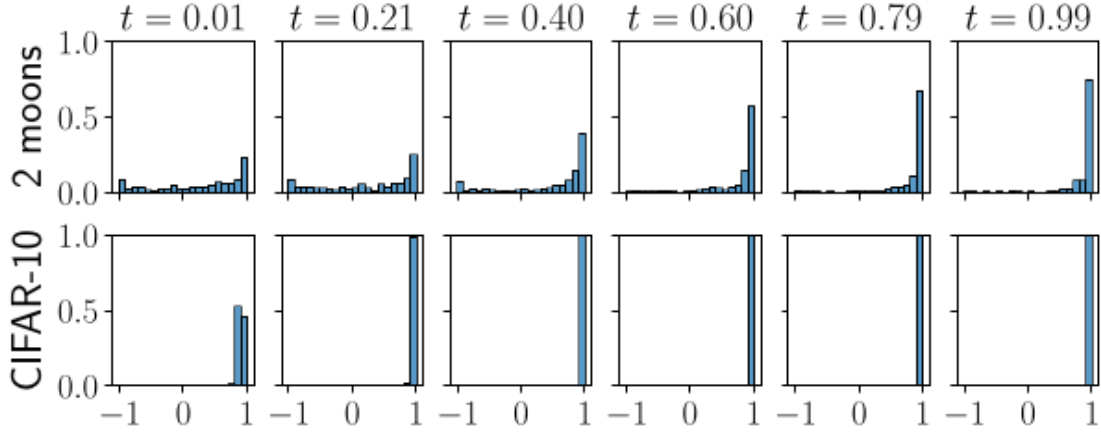


Figure 1: The histograms of the cosine similarities between $u_t((1-t)x_0 + tx_1)$ and $u_t((1-t)x_0 + tx_1 \mid x_1) = x_1 - x_0$ are displayed for various time values t and two datasets: 2-moons and CIFAR-10. (Bertrand et al. 2025, fig. 1)

CIFAR-10 Preprocessing and Statistics

We use standard CIFAR-10 preprocessing following the [conditional-flow-matching](#) repository:

```
dataset = datasets.CIFAR10(
    root="./data",
    train=True,
    download=True,
    transform=transforms.Compose([
        transforms.ToTensor(),
        transforms.Normalize((0.5, 0.5, 0.5), (0.5, 0.5, 0.5)),
    ]),
)
```

With `ToTensor()` (pixels in $[0, 1]$) followed by normalization:

$$x = \frac{\text{pixel} - 0.5}{0.5} = 2 \cdot \text{pixel} - 1 \in [-1, 1]$$

Each coordinate is bounded and roughly centered near 0. In high dimension, the vector behaves like a collection of weakly dependent coordinates.

Key Parameters

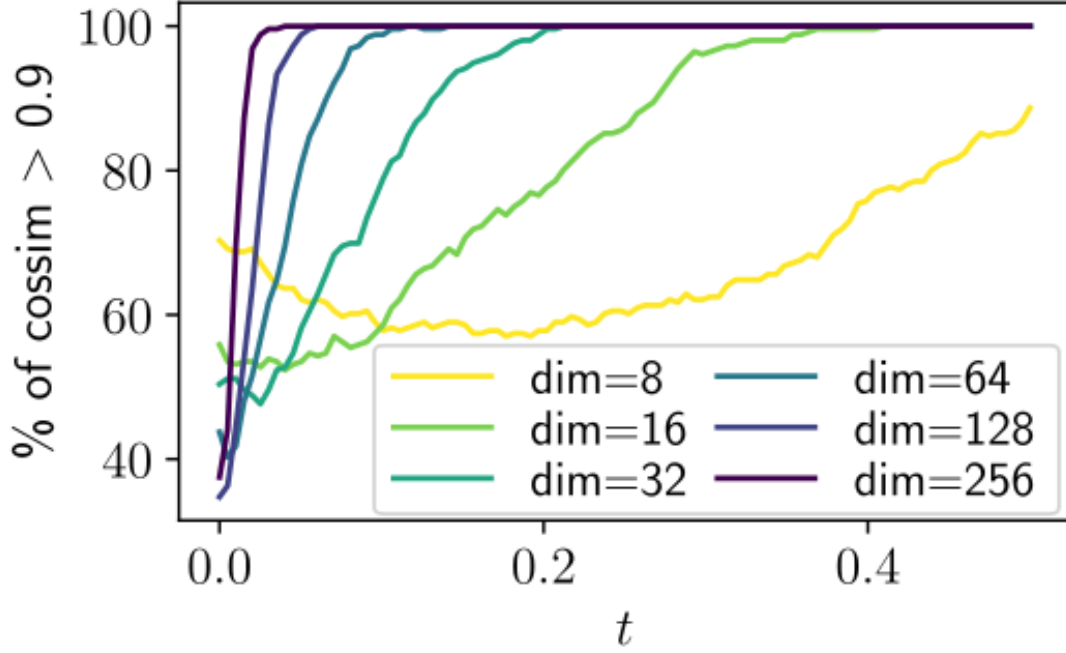


Figure 2: Proportion of samples x_t for which the cosine similarity between $u_t(x_t)$ and $u_t(x_t | x_1)$ exceeds 0.9, as a function of time t . This analysis is performed across multiple spatial resolutions of the Imagenette dataset, obtaining $d \times d$ images by spatial subsampling. (Bertrand et al. 2025, fig. 2)

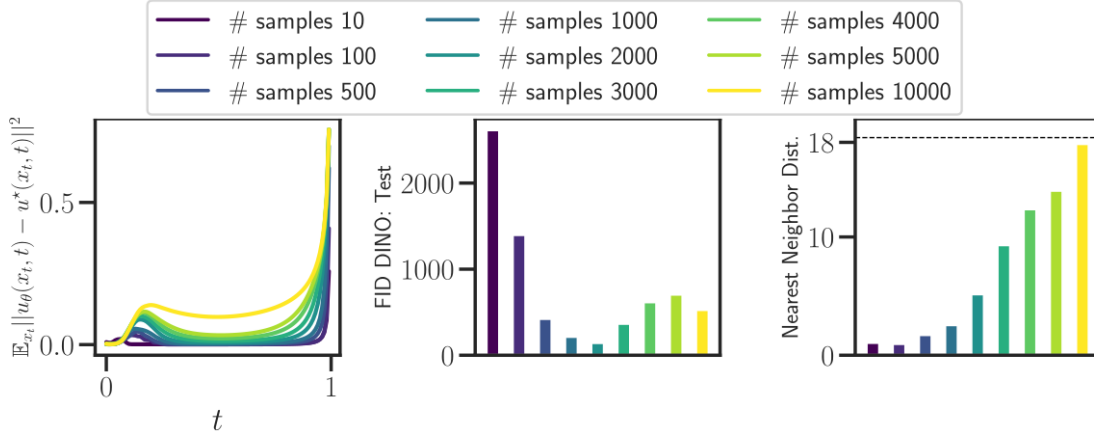


Figure 3: Failure to learn the marginal velocity field on CIFAR-10: the leftmost figure represents the average error between the marginal empirical velocity field u_t and the learned velocity u_t^θ for multiple values of time t . All the quantities are computed/learned on a varying number of training samples (10 to 10^4) of the CIFAR-10 dataset. We are not interested in the other figures in this blog post. (Bertrand et al. 2025, fig. 3)

Parameter	Value	Description
d	$3 \times 32 \times 32 = 3072$	Dimension
σ^2	≈ 0.24	Per-coordinate variance
n	50,000	Number of training samples

A [commonly cited computation](#) over the CIFAR-10 training set gives:

- mean $\approx (0.4914, 0.4822, 0.4465)$
- std $\approx (0.247, 0.243, 0.261)$

(Those are for pixel values in $[0, 1]$, i.e., after `ToTensor()`.)

With normalization by Torchvision, the variance transforms as $y = \frac{x - \mu}{\sigma}$ with $\mu = 0.5$ and $\sigma = 0.5$, so:

$$\text{Var}(y) = \frac{\text{Var}(x)}{\sigma^2} = \frac{\text{Var}(x)}{0.25} = 4\text{Var}(x).$$

Using the CIFAR-10 channel stds above:

- $\text{Var}(x_R) \approx 0.247^2 \approx 0.0610 \Rightarrow \text{Var}(y_R) \approx 4 \cdot 0.0610 \approx 0.244$
- $\text{Var}(x_G) \approx 0.243^2 \approx 0.0590 \Rightarrow \text{Var}(y_G) \approx 0.236$

- $\text{Var}(x_B) \approx 0.261^2 \approx 0.0681 \Rightarrow \text{Var}(y_B) \approx 0.273$

These are all of the same order, and for simplicity we use a representative value

$$\sigma^2 \approx 0.24$$

for the per-coordinate variance of the normalized dataset.

Remark: Channel Statistics vs. Pixel Statistics

It is natural to ask how **channel-wise statistics** (mean and variance computed over all pixels in a given channel) relate to **per-pixel statistics** (mean and variance at a fixed spatial location).

Fix a channel (say, red) and let $X_{u,v}$ denote the intensity at pixel location (u, v) for a random CIFAR-10 image. Let (U, V) be a pixel location drawn uniformly at random from the grid, independently of the image, and define the “global random pixel”

$$Y := X_{U,V}.$$

By the law of total variance,

$$\text{Var}(Y) = \mathbb{E}_{U,V}[\text{Var}(X_{u,v})] + \text{Var}_{U,V}(\mathbb{E}[X_{u,v}]).$$

Interpreting the terms:

- $\mathbb{E}_{U,V}[\text{Var}(X_{u,v})]$ is the **average per-pixel variance** (variance across images, averaged over spatial locations).
- $\text{Var}(Y)$ is the **global channel variance** when pooling all pixels in that channel.
- The nonnegative term $\text{Var}_{U,V}(\mathbb{E}[X_{u,v}])$ measures how much the **mean image** varies across spatial location.

Therefore,

$$\mathbb{E}_{U,V}[\text{Var}(X_{u,v})] \leq \text{Var}(Y),$$

and the gap is exactly the “variance of the per-pixel means.”

On CIFAR-10, this spatial variation term is present (e.g., backgrounds vs. object centers) but is not typically huge compared to the intrinsic per-pixel variability. Hence:

- channel-level variances (like the CIFAR-10 stds above) and
- average per-pixel variances

are of the **same order of magnitude**, with the global (pooled) variance usually slightly larger.

Gaussian Toy Model and Its Limitations

In the theoretical analysis below, we **idealize** the dataset as consisting of i.i.d. Gaussian vectors:

$$x_1, \dots, x_n \in \mathbb{R}^d, \quad x_i \sim \mathcal{N}(0, \sigma^2 I_d),$$

with

$$d = 3072, \quad n = 50,000, \quad \sigma^2 \approx 0.24.$$

This model should be understood as a **toy approximation**: it preserves only very coarse statistics (dimension, scale of per-coordinate variance, approximate centering) and treats coordinates as independent and identically distributed.

Of course, real CIFAR-10 images are far from i.i.d. Gaussian vectors. The true covariance has a highly nontrivial structure, as illustrated by the empirical eigenvalue spectrum of the covariance matrix Figure 4.

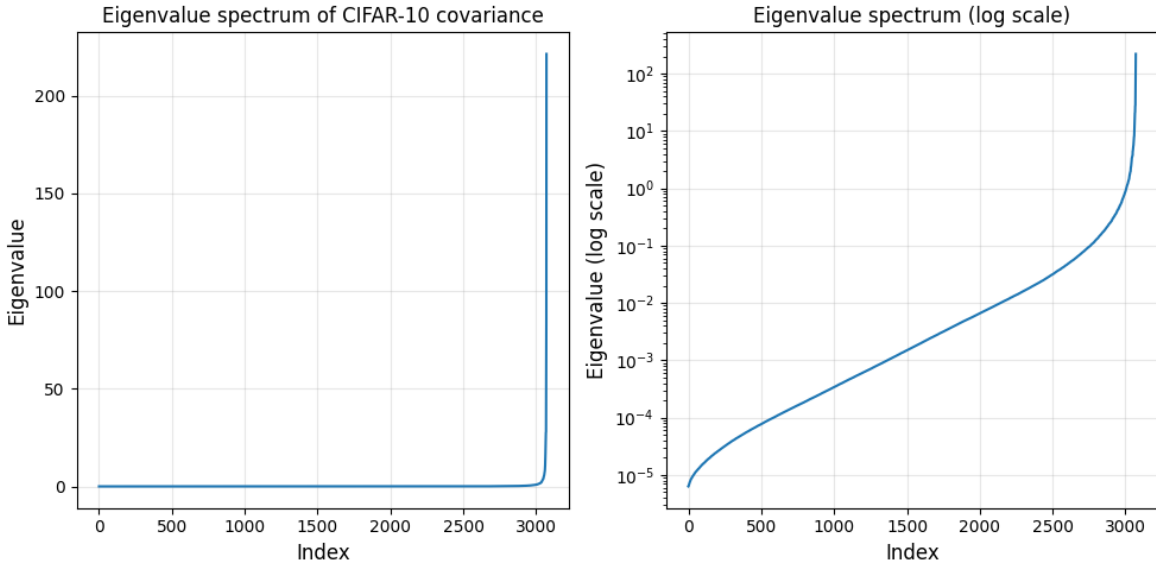


Figure 4: The eigenvalues of the covariance matrix of the CIFAR-10 dataset.

The pronounced decay and variability of eigenvalues in Figure 4 reflect strong dependencies across pixels and channels (e.g., spatial correlation, global color structure, object/background patterns). The i.i.d. Gaussian model thus serves only as a **simplified reference**, not as a realistic generative model for CIFAR-10.

Marginal Velocity Field Collapse: A Statistical Physics Perspective

This section provides a rigorous analysis of the collapse phenomenon in the marginal velocity field using tools from statistical physics, specifically the Random Energy Model (REM) and Large Deviation Principles (LDP).

Gibbs Measure and Energy Formulation

The Marginal Velocity Field as a Gibbs Measure

The marginal velocity field in Flow Matching is given by:

$$u_t(x) = \sum_{i=1}^n \lambda_i(x, t) \frac{x^{(i)} - x}{1 - t}$$

where the weights $\lambda_i(x, t)$ take the form:

$$\lambda_i(x, t) = \frac{\exp\left(-\frac{\|x - tx^{(i)}\|^2}{2(1-t)^2}\right)}{\sum_{j=1}^n \exp\left(-\frac{\|x - tx^{(j)}\|^2}{2(1-t)^2}\right)}.$$

This is precisely a **Boltzmann distribution** at inverse temperature $\beta = 1$ over “states” $i \in \{1, \dots, n\}$ with energy:

$$E_i(x, t) = \frac{\|x - tx^{(i)}\|^2}{2(1-t)^2}.$$

The denominator defines the **partition function**:

$$Z(x, t) = \sum_{j=1}^n e^{-E_j(x, t)}.$$

Assumptions on the Data

From Section , we consider a **Gaussian proxy** for the training data:

$$x^{(i)} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2 I_d), \quad i = 1, \dots, n.$$

This approximation captures the essential high-dimensional geometry while enabling analytical tractability. For CIFAR-10 after standard normalization, $\sigma^2 \approx 0.24$ provides a reasonable proxy.

The “Planted” Trajectory

Consider a trajectory starting from noise and targeting a specific training image:

$$x_t = (1-t)x_0 + tx_1, \quad x_0 \sim \mathcal{N}(0, I_d), \quad x_1 = x^{(1)}.$$

We call $i = 1$ the **special (planted) index**. For this index:

$$x_t - tx^{(1)} = (1-t)x_0$$

which gives the **planted energy**:

$$E_1(x_t, t) = \frac{\|(1-t)x_0\|^2}{2(1-t)^2} = \frac{\|x_0\|^2}{2}.$$

By the law of large numbers for chi-squared random variables:

$$\frac{E_1}{d} \xrightarrow{d \rightarrow \infty} \frac{1}{2} \quad \text{almost surely.}$$

Hence the planted contribution to the partition function is:

$$Z_1 = e^{-E_1} \approx e^{-d/2}.$$

The “Bulk” (REM-like) Energies

For $j \neq 1$, we have:

$$x_t - tx^{(j)} = (1-t)x_0 + t(x_1 - x^{(j)}).$$

Under the Gaussian proxy, $x_1 - x^{(j)} \sim \mathcal{N}(0, 2\sigma^2 I_d)$ (independent of x_0). Therefore:

$$x_t - tx^{(j)} \sim \mathcal{N}(0, s^2(t)I_d), \quad s^2(t) := (1-t)^2 + 2\sigma^2 t^2.$$

The bulk energies become:

$$E_j(x_t, t) = \frac{\|x_t - tx^{(j)}\|^2}{2(1-t)^2} \approx \frac{s^2(t)}{2(1-t)^2} \cdot \chi_d^2.$$

Define the **time-dependent energy scale**:

$$c(t) = \frac{s^2(t)}{2(1-t)^2} = \frac{1}{2} \left(1 + \frac{2\sigma^2 t^2}{(1-t)^2} \right).$$

Key observation: The bulk energies are approximately i.i.d. across $j \geq 2$, each extensive (order d). This is the classic setup where **Random Energy Model (REM)** tools apply.

We define the partition function associated with the bulk states as:

$$Z_{\text{bulk}}(x, t) := \sum_{j=2}^n e^{-E_j(x, t)}.$$

Following the statistical-physics convention, we call the normalized log of the bulk partition function the (quenched) **bulk free-energy density** at inverse temperature $\beta = 1$:

$$\Phi(1) := \lim_{d \rightarrow \infty} \frac{1}{d} \log Z_{\text{bulk}}(x_t, t),$$

whenever the limit exists. When we want to keep the time dependence explicit we will also write $\Phi(1, t)$.

Collapsed vs Uncollapsed States (Glass vs Liquid)

Definition of Collapse

The central question is whether the partition function $Z(x_t, t)$ is dominated by:

- **One special term** (the planted state), or
- **Exponentially many comparable terms** (the bulk).

This leads to two phases:

Phase	Statistical Physics	Flow Matching	Behavior
Collapsed	Glass / Condensed	Memorization	$\lambda_1 \approx 1$
Uncollapsed	Liquid	Generalization	Mass spread over many j 's

Glass Phase (Collapsed / Memorization)

In the collapsed phase, $Z \approx Z_1 = e^{-E_1}$, meaning $\lambda_1 \approx 1$.

The marginal velocity field then reduces to:

$$u_t(x_t) \approx \frac{x^{(1)} - x_t}{1 - t} = \frac{x_1 - x_t}{1 - t} = x_1 - x_0.$$

This is precisely the **conditional velocity field**. The marginal flow aligns perfectly with the teacher trajectory, leading to memorization of training data.

Liquid Phase (Uncollapsed / Generalization)

In the liquid phase, $Z \approx Z_{\text{bulk}} = \sum_{j=2}^n e^{-E_j}$, with mass distributed across many states.

The velocity field is a genuine mixture of directions toward different training points, enabling interpolation and generalization.

The Phase Transition

The transition between phases depends on:

- **Energy advantage** of the planted state (order d)
- **Entropy** from having n competitors (order $\log n$)

The competition is controlled by the **dimensionless ratio**:

$$\alpha = \frac{\log n}{d}.$$

For typical datasets (e.g., CIFAR-10 with $d = 3072$, $n = 50,000$):

$$\alpha = \frac{\log(50,000)}{3072} \approx 0.0035 \ll 1.$$

This tiny α means the entropy budget is negligible compared to extensive energy gaps, placing the system deep in the collapsed regime for most values of t .

Simple Heuristic Analysis

Typical Energy Gap

Compare a typical bulk energy to the planted energy. The **typical energy gap** is:

$$E_j - E_1 \approx \frac{d}{2} \left(\frac{s^2(t)}{(1-t)^2} - 1 \right) = \frac{d}{2} \cdot \frac{2\sigma^2 t^2}{(1-t)^2} = \frac{d\sigma^2 t^2}{(1-t)^2}.$$

Therefore, the log-weight ratio is:

$$\log \frac{\lambda_j}{\lambda_1} \approx -(E_j - E_1) \approx -\frac{d\sigma^2 t^2}{(1-t)^2}.$$

This shows **exponential-in- d suppression** of bulk weights relative to the planted weight.

Crude Annealed Entropy Estimate

A naive estimate of the bulk partition function uses the **annealed approximation**:

$$Z_{\text{bulk}} \approx (n-1) \cdot \mathbb{E}[e^{-E_j}].$$

However, this is often wrong because e^{-E_j} has exponentially broad fluctuations. The sum is dominated by **rare low-energy states**, not typical ones. This is precisely where REM physics becomes essential.

Energy vs Entropy Competition

The competition can be summarized as:

- **Planted contribution:** $Z_1 \sim e^{-d/2}$
- **Bulk contribution (naive):** $Z_{\text{bulk}} \sim n \cdot e^{-d \cdot c(t)} = e^{\alpha d - d \cdot c(t)}$

For the planted state to dominate, we need the energy advantage to exceed the entropy:

$$\frac{d\sigma^2 t^2}{(1-t)^2} \gg \log n.$$

Rearranging, collapse holds whenever:

$$\frac{t}{1-t} \gg \sqrt{\frac{\log n}{d\sigma^2}}.$$

For CIFAR-like numbers, the right-hand side is tiny (≈ 0.12), so collapse can occur at small t .

Simple Collapse Time Estimate

Setting the two sides equal gives a heuristic collapse time:

$$t_C^{\text{heuristic}} \approx \frac{\sqrt{\alpha/\sigma^2}}{1 + \sqrt{\alpha/\sigma^2}}.$$

For CIFAR-10 ($\alpha \approx 0.0035$, $\sigma^2 \approx 0.24$):

$$t_C^{\text{heuristic}} \approx 0.11.$$

Caveat: This estimate compares typical bulk energy to planted energy. A rigorous analysis must consider the **best** competitor among $n-1$ states, not a typical one. This requires REM/LDP tools.

Rigorous REM/LDP Derivation

Large Deviation Principles: Foundations

Why LDP?

The Law of Large Numbers tells us *where* random variables concentrate, but not *how unlikely* deviations are. Large Deviation Theory provides **exponential decay rates** for rare events.

Definition

A sequence of random variables X_d satisfies a **Large Deviation Principle (LDP)** with rate function $I(x)$ if:

$$\mathbb{P}(X_d \approx x) \asymp e^{-dI(x)}$$

where:

- d is the large parameter (dimension)
- $I(x) \geq 0$ with $I(x) = 0$ at the typical value
- Larger $I(x)$ means exponentially rarer events

Cramér's Theorem

For i.i.d. random variables $X_1, \dots, X_d$ with finite moment generating function, the empirical average $S_d = \frac{1}{d} \sum_{k=1}^d X_k$ satisfies an LDP:

$$\mathbb{P}(S_d \approx x) \asymp e^{-dI(x)}$$

where the **rate function** is the Legendre transform of the log-MGF:

$$I(x) = \sup_{\lambda \in \mathbb{R}} (\lambda x - \log \mathbb{E}[e^{\lambda X_1}]).$$

LDP for Chi-Squared (Energy Density)

Deriving the Rate Function

Let $Z_k \sim \mathcal{N}(0, 1)$ i.i.d., so $\chi_d^2 = \sum_{k=1}^d Z_k^2$. Define:

$$Y_d = \frac{\chi_d^2}{d} = \frac{1}{d} \sum_{k=1}^d Z_k^2.$$

For $X = Z^2$, the log-MGF is:

$$\log \mathbb{E}[e^{\lambda Z^2}] = -\frac{1}{2} \log(1 - 2\lambda), \quad \lambda < \frac{1}{2}.$$

Applying Cramér's theorem and computing the Legendre transform:

$$I_\chi(y) = \sup_{\lambda < 1/2} \left(\lambda y + \frac{1}{2} \log(1 - 2\lambda) \right) = \frac{1}{2}(y - 1 - \log y), \quad y > 0.$$

Key properties:

- Minimum at $y = 1$ with $I_\chi(1) = 0$ (typical value)
- Convex and non-negative
- Deviations are exponentially unlikely in d

Energy Density LDP

For bulk energies with $E = c(t) \cdot \chi_d^2$, the energy density $e = E/d = c \cdot Y_d$ satisfies:

$$\mathbb{P}(e \approx \varepsilon) \asymp e^{-d I(\varepsilon)}$$

where:

$$I(\varepsilon) = \frac{1}{2} \left(\frac{\varepsilon}{c} - 1 - \log \frac{\varepsilon}{c} \right)$$

This is the **rigorous backbone** for all REM arguments.

Bulk Free Energy via Laplace Principle

Complexity (Entropy of States)

With $n = e^{\alpha d}$ bulk energies, the expected **number of states** with energy density near ε is:

$$\#(\varepsilon) \approx n \cdot \mathbb{P}(e \approx \varepsilon) \approx \exp\{d(\alpha - I(\varepsilon))\}.$$

The quantity:

$$\Sigma(\varepsilon) = \alpha - I(\varepsilon)$$

is the **complexity** (entropy density of states at energy ε).

Laplace Principle

The bulk partition function is approximately:

$$Z_{\text{bulk}} = \sum_{j=2}^n e^{-E_j} \approx \int d\varepsilon \exp\{d(\Sigma(\varepsilon) - \varepsilon)\}.$$

Using the LDP for the bulk energies, the bulk free-energy density $\Phi(1)$ defined above can be computed via a Laplace (saddle-point) principle:

$$\Phi(1) = \sup_{\varepsilon > 0} [\alpha - I(\varepsilon) - \varepsilon]$$

Glass vs Liquid Phase Transition

The Variational Problem

Define $F(\varepsilon) = \alpha - I(\varepsilon) - \varepsilon$. The maximizer determines the phase.

Liquid Phase (Interior Solution)

If the maximizer is interior, it satisfies $F'(\varepsilon) = 0$, i.e., $I'(\varepsilon) = -1$.

Computing:

$$I'(\varepsilon) = \frac{1}{2} \left(\frac{1}{c} - \frac{1}{\varepsilon} \right) = -1$$

yields the **liquid saddle point**:

$$\varepsilon_* = \frac{c}{1 + 2c}$$

This solution is valid only if there are exponentially many states there: $\Sigma(\varepsilon_*) = \alpha - I(\varepsilon_*) \geq 0$.

Glass Phase (Boundary Solution)

If $\alpha - I(\varepsilon_*) < 0$, there are too few states at the liquid saddle. The supremum hits the boundary where complexity vanishes:

$$I(\varepsilon_g) = \alpha$$

The partition sum is then dominated by the **lowest-energy extremes**: this is the glass/condensed phase.

Freezing Threshold

The critical entropy level separating phases is:

$$\alpha_c(c) = I(\varepsilon_*) = \frac{1}{2} \left(\log(1 + 2c) - \frac{2c}{1 + 2c} \right)$$

Phase diagram:

- **Liquid phase:** $\alpha \geq \alpha_c(c)$, maximizer at ε_* , free energy $\Phi(1) = \alpha - I(\varepsilon_*) - \varepsilon_*$
- **Glass phase:** $\alpha < \alpha_c(c)$, maximizer at ε_g with $I(\varepsilon_g) = \alpha$, free energy $\Phi(1) = -\varepsilon_g$

Collapse Criterion from Free Energy

Full Partition Function

The complete partition function is:

$$Z = e^{-E_1} + Z_{\text{bulk}}, \quad \lambda_1 = \frac{e^{-E_1}}{Z}.$$

Since $E_1/d \rightarrow 1/2$, we have:

$$\frac{1}{d} \log \frac{Z_{\text{bulk}}}{e^{-E_1}} \rightarrow \Phi(1) + \frac{1}{2}.$$

Collapse Condition

$$\lambda_1 \rightarrow 1 \iff \Phi(1) < -\frac{1}{2}$$

More precisely, for large d :

$$\lambda_1 \approx \frac{1}{1 + \exp\{d(\Phi(1) + 1/2)\}}.$$

Glass Phase Collapse Criterion

In the glass phase, $\Phi(1) = -\varepsilon_g$, so collapse requires:

$$-\varepsilon_g < -\frac{1}{2} \iff \varepsilon_g > \frac{1}{2}$$

where ε_g is defined implicitly by $I(\varepsilon_g) = \alpha$.

Interpretation: Collapse occurs when even the *best* spurious competitor (with energy density ε_g) has higher energy than the planted state (energy density $1/2$).

Liquid Phase Collapse Criterion

In the liquid phase, collapse requires:

$$\alpha - I(\varepsilon_*) - \varepsilon_* < -\frac{1}{2}.$$

This gives an explicit inequality in c and α .

Explicit Collapse Time

Collapse Time Equation

The collapse time t_C is defined by $\varepsilon_g(t_C) = 1/2$. Setting $\varepsilon_g = 1/2$ in $I(\varepsilon_g) = \alpha$:

$$\alpha = I(1/2) = \frac{1}{2} \left(\frac{1/2}{c} - 1 - \log \frac{1/2}{c} \right) = \frac{1}{2} \left(\frac{1}{2c} - 1 + \log(2c) \right).$$

This gives the **REM-rigorous collapse boundary**:

$$\alpha = \frac{1}{2} \left(\log(2c(t_C)) + \frac{1}{2c(t_C)} - 1 \right)$$

with $c(t) = \frac{1}{2} \left(1 + \frac{2\sigma^2 t^2}{(1-t)^2} \right)$.

CIFAR-10 Proxy Calculation

For CIFAR-10: $d = 3072$, $n = 50,000$, $\sigma^2 = 0.24$.

Compute:

$$\alpha = \frac{\log(50,000)}{3072} \approx 0.003522.$$

Solving the collapse equation yields:

$$t_C \approx 0.341$$

with corresponding $c(t_C) \approx 0.5644$.

Interpretation: Under the i.i.d. Gaussian proxy + REM idealization, for $t \lesssim 0.34$, the planted datapoint is no longer guaranteed to dominate because among n training points, one can typically find at least one spurious point whose energy matches or beats the planted energy. For $t \gtrsim 0.34$, collapse occurs and the planted state dominates.

Why CIFAR is Always in the Glass Phase

For CIFAR-like $\alpha \ll 1$, we have $\alpha \ll \alpha_c(c(t))$ for all $t \in [0, 1)$. The system is **always in the glass phase**, meaning that the bulk partition function is dominated by a few extreme low-energy states.

Summary of Key Equations

Bulk Energy LDP

$$\mathbb{P}\left(\frac{E}{d} \approx \varepsilon\right) \asymp e^{-dI(\varepsilon)}, \quad I(\varepsilon) = \frac{1}{2} \left(\frac{\varepsilon}{c} - 1 - \log \frac{\varepsilon}{c} \right)$$

Bulk Quenched Free Energy

$$\Phi(1) = \sup_{\varepsilon > 0} [\alpha - I(\varepsilon) - \varepsilon], \quad \alpha = \frac{\log n}{d}$$

Glass vs Liquid Threshold

$$\alpha_c(c) = \frac{1}{2} \left(\log(1 + 2c) - \frac{2c}{1 + 2c} \right)$$

Collapse Criterion

$$\lambda_1 \rightarrow 1 \iff \Phi(1) < -\frac{1}{2}$$

In the glass phase ($\alpha < \alpha_c$):

$$\lambda_1 \rightarrow 1 \iff \varepsilon_g > \frac{1}{2}, \quad I(\varepsilon_g) = \alpha$$

Collapse Time (Glass Phase)

$$\alpha = \frac{1}{2} \left(\log(2c(t_C)) + \frac{1}{2c(t_C)} - 1 \right)$$

Extension to Non-Isotropic Covariance

Motivation

The isotropic Gaussian proxy $x^{(i)} \sim \mathcal{N}(0, \sigma^2 I_d)$ is convenient but unrealistic. Real image datasets have structured covariance with varying eigenvalues (some directions have high variance, others low). The REM/LDP framework extends naturally to non-isotropic covariance. The structure of the theory remains identical — only the **rate function** $I(\varepsilon)$ changes.

Setup: General Covariance Model

Assume training images follow:

$$x^{(i)} \sim \mathcal{N}(0, C), \quad i = 1, \dots, n$$

where C is a $d \times d$ positive definite covariance matrix with eigendecomposition:

$$C = U \text{diag}(\mu_1, \dots, \mu_d) U^\top.$$

Deriving the Effective Covariance

Recall the bulk energy for $j \geq 2$:

$$E_j = \frac{\|x_t - tx^{(j)}\|^2}{2(1-t)^2}, \quad x_t = (1-t)x_0 + tx_1.$$

Step 1: Decompose the displacement

$$x_t - tx^{(j)} = (1-t)x_0 + t(x_1 - x^{(j)}).$$

Step 2: Identify the distributions

- $x_0 \sim \mathcal{N}(0, I_d)$ (noise)
- $x_1 - x^{(j)} \sim \mathcal{N}(0, 2C)$ (difference of two i.i.d. samples from $\mathcal{N}(0, C)$)

Step 3: Compute the covariance of $x_t - tx^{(j)}$

Since x_0 and $(x_1 - x^{(j)})$ are independent:

$$\text{Cov}(x_t - tx^{(j)}) = (1-t)^2 I_d + 2t^2 C =: \Sigma_t.$$

Step 4: Define the effective covariance for the energy

The energy involves $\|x_t - tx^{(j)}\|^2/(2(1-t)^2)$. Let $g \sim \mathcal{N}(0, I_d)$ be a standard Gaussian vector. Then:

$$x_t - tx^{(j)} \stackrel{d}{=} \Sigma_t^{1/2} g$$

and the energy becomes:

$$E_j = \frac{g^\top \Sigma_t g}{2(1-t)^2} = \frac{1}{2} g^\top C_t g$$

where the **effective covariance** is:

$$C_t = \frac{\Sigma_t}{(1-t)^2} = I_d + \frac{2t^2}{(1-t)^2} C.$$

Eigenvalues of the Effective Covariance

If C has eigenvalues $\{\mu_k\}_{k=1}^d$, then C_t has eigenvalues:

$$\lambda_k^{(t)} = 1 + \frac{2t^2}{(1-t)^2} \mu_k, \quad k = 1, \dots, d.$$

The bulk energy density is:

$$\frac{E_j}{d} = \frac{1}{2d} \sum_{k=1}^d \lambda_k^{(t)} Z_k^2$$

where $Z_k \sim \mathcal{N}(0, 1)$ are i.i.d. (the components of g in the eigenbasis of C_t).

Recovery of isotropic case: When $C = \sigma^2 I_d$, all $\mu_k = \sigma^2$, so:

$$\lambda_k^{(t)} = 1 + \frac{2\sigma^2 t^2}{(1-t)^2} = 2c(t)$$

which matches our earlier formula (the factor of 2 accounts for the $1/2$ in $E_j = \frac{1}{2} g^\top C_t g$).

LDP for Weighted Chi-Squared: Gärtner-Ellis Theorem

In the general covariance case, the energy density is a weighted sum of chi-squared variables:

$$\frac{E_j}{d} = \frac{1}{2d} \sum_{k=1}^d \lambda_k^{(t)} Z_k^2,$$

a sum of independent but **non-identically distributed** terms (the weights $\lambda_k^{(t)}$ vary with k). To obtain an LDP in this heterogeneous setting, we use the **Gärtner–Ellis theorem**, which handles such triangular arrays via the limit of the log-moment generating function.

Step 1: Compute the scaled cumulant generating function

For a single term λZ^2 with $Z \sim \mathcal{N}(0, 1)$:

$$\mathbb{E}[e^{\theta \lambda Z^2}] = \frac{1}{\sqrt{1 - 2\theta\lambda}}, \quad \theta < \frac{1}{2\lambda}.$$

Thus:

$$\log \mathbb{E}[e^{\theta \lambda Z^2}] = -\frac{1}{2} \log(1 - 2\theta\lambda).$$

Step 2: Sum over all eigenvalues

The scaled log-MGF for the energy density $e = E_j/d$ is:

$$\Lambda_t(\theta) = \lim_{d \rightarrow \infty} \frac{1}{d} \log \mathbb{E} \left[e^{d\theta \cdot \frac{1}{2d} \sum_k \lambda_k^{(t)} Z_k^2} \right] = \lim_{d \rightarrow \infty} \frac{1}{d} \sum_{k=1}^d \left(-\frac{1}{2} \log(1 - \theta \lambda_k^{(t)}) \right).$$

This limit exists if the **empirical spectral distribution** of C_t converges:

$$\Lambda_t(\theta) = \int \left(-\frac{1}{2} \log(1 - \theta\lambda) \right) d\nu_t(\lambda)$$

where ν_t is the limiting spectral measure of $\{\lambda_k^{(t)}\}$.

Step 3: Legendre transform gives the rate function

By the Gärtner-Ellis theorem:

$$I_t(\varepsilon) = \sup_{\theta < 1/\lambda_{\max}^{(t)}} (\theta\varepsilon - \Lambda_t(\theta))$$

This is the **drop-in replacement** for the isotropic rate function.

The Full Pipeline with General Covariance

The REM framework carries through unchanged:

1. Bulk free energy:

$$\Phi(1, t) = \sup_{\varepsilon > 0} [\alpha - I_t(\varepsilon) - \varepsilon]$$

2. Collapse criterion:

$$\lambda_1 \rightarrow 1 \iff \Phi(1, t) < -\frac{1}{2}$$

3. Planted weight approximation:

$$\lambda_1(t) \approx \frac{1}{1 + \exp\{d(\Phi(1, t) + 1/2)\}}$$

Only $I_t(\varepsilon)$ changes — the rest of the theory is identical.

Error Analysis: Training Difficulties Across Time

As observed in Figure 3, the learned velocity field u_t^θ exhibits a characteristic error profile when approximating the marginal velocity field u_t : low error near $t = 0$, a peak around $t \approx 0.1$, a decrease for intermediate t , and a final spike near $t = 1$. This section tries to explain these four regimes.

Training Error Trajectory: Four Regimes

Regime 1: $t \approx 0$ — Trivial Learning Phase

For small t , the interpolant is dominated by noise: $X_t = (1 - t)X_0 + tX_1 \approx X_0$. The marginal velocity field at X_t is:

$$u_t(X_t) = \frac{\mathbb{E}[X_1 | X_t] - X_t}{1 - t}.$$

When $t \approx 0$, the interpolant $X_t \approx X_0$ carries almost no information about X_1 . Under the empirical distribution, the posterior mean satisfies:

$$\mathbb{E}[X_1 | X_t] \approx \bar{x} \approx 0$$

where $\bar{x}$ is the mean of the training data (approximately zero after normalization).

Therefore, the optimal velocity field simplifies to:

$$u_t(X_t) \approx \frac{0 - X_t}{1 - t} \approx \frac{-X_t}{1} = -X_t.$$

This is essentially a **negation map**: the neural network only needs to learn $u_t^\theta(x) \approx -x$, which is trivial.

Regime 2: $t \approx t_C$ — Peak Difficulty (Collapse Transition)

The error peaks around $t \approx 0.1$ – 0.2 , which corresponds to the approach toward the **collapse transition** at t_C .

In this regime, the posterior $p(X_1 | X_t)$ is **neither uniform nor concentrated**:

- The interpolant X_t now contains some information about X_1 , so the posterior is no longer uniform over training points.
- But collapse has not yet occurred, so multiple training points still have significant posterior weight.

This creates several sources of difficulty:

1. **Complex posterior structure:** The neural network must learn to predict $\mathbb{E}[X_1 | X_t]$, which is a weighted average over multiple training points. The weights $\lambda_i(X_t, t)$ vary rapidly with X_t .
2. **Signal-to-noise transition:** The interpolant $X_t = (1-t)X_0 + tX_1$ is transitioning from noise-dominated to signal-dominated. The network must extract X_1 from X_t , but X_t is still far from X_1 .
3. **Sharp phase transition:** The REM analysis shows that the energy gaps scale as:

$$E_j - E_1 \approx \frac{d\sigma^2 t^2}{(1-t)^2}.$$

Near t_C , the system transitions from having exponentially many competing states to a single dominant state. This transition is sharp in high dimensions, creating rapid variation in the velocity field as a function of both x and t .

4. **High sensitivity:** Near the transition, the map $x \mapsto \arg \max_i \lambda_i(x, t)$ is extremely sensitive to perturbations. Small changes in X_t can cause different training points to dominate the posterior.

Regime 3: $t > t_C$ — Decreasing Error (Concentration)

After the collapse transition ($t > t_C$), the error **decreases**. The posterior concentrates on the planted index with $\lambda_1(X_t, t) \approx 1$.

In this regime:

$$u_t(X_t) \approx \frac{x^{(1)} - X_t}{1-t} = \frac{X_1 - X_t}{1-t} = X_1 - X_0.$$

The marginal velocity field becomes essentially equal to the **conditional velocity field**. The neural network's task simplifies: given X_t , it must predict $X_1 - X_0$, but now X_t strongly determines which X_1 generated it.

The learning problem has two favorable properties:

1. **Low posterior uncertainty:** With $\lambda_1 \approx 1$, there is little ambiguity about which training point generated X_t .
2. **Increasing signal:** As t grows, X_t becomes closer to X_1 , making it easier to infer the target direction $X_1 - X_0$.

Geometrically, the velocity field becomes piecewise-affine: within each Voronoi cell (determined by proximity to training points), the network must approximate

$$\frac{x^{(i)} - x}{1 - t},$$

where i indexes the training point whose cell contains x . Although the coefficient $1/(1 - t)$ grows as t increases, the key simplification is that the posterior has collapsed—the network no longer faces the challenge of averaging over multiple competing directions.

Regime 4: $t \rightarrow 1$ — Final Blow-Up

Near $t = 1$, the error spikes again due to **Voronoi discontinuities**.

Voronoi Discontinuities (Nearest-Neighbor Regime)

As $t \rightarrow 1$, the softmax temperature $(1 - t)^2 \rightarrow 0$ drives the weights toward a **one-hot distribution**:

$$\lambda_i(x, t) \xrightarrow{t \rightarrow 1} \mathbf{1} \left[i = \arg \min_j \|x - x^{(j)}\| \right].$$

The marginal velocity field approaches:

$$u_t(x) \approx \frac{x^{(i^*)} - x}{1 - t}, \quad \text{where } i^* = \arg \min_j \|x - x^{(j)}\|.$$

This is a **piecewise-affine vector field** that switches abruptly across **Voronoi cell boundaries** in $\mathbb{R}^d$. Such discontinuous fields are notoriously difficult to approximate with smooth neural networks, explaining the error spike near $t \approx 1$.

Summary: Error Profile Across Time

Regime	Time	Error	Explanation
1	$t \approx 0$	Low	Posterior mean ≈ 0 ; target is $\approx -X_0$ (trivial)
2	$t \approx t_C$	Peak	Complex posterior; phase transition; high sensitivity
3	$t > t_C$	Decreasing	Posterior concentrates; clear signal; simpler target structure
4	$t \rightarrow 1$	Spike	Voronoi discontinuities; $1/(1 - t)$ blow-up

CIFAR-10: Numerical Analysis

Let us briefly review the framework of flow matching. In flow matching, we aim to learn a velocity field $u_t(x)$ that transports samples from a noise distribution to the data distribution. A central phenomenon in this process is **velocity field collapse**: at a certain critical time t_C , the marginal velocity field becomes dominated by a single training example, effectively making it singular.

Key question: When does this collapse happen?

The timing of the collapse is determined by a fundamental parameter:

$$\alpha = \frac{\log n}{d}$$

where n is the number of training samples and d is the data dimension. For CIFAR-10, $n = 50,000$ and $d = 3 \times 32 \times 32 = 3072$, yielding $\alpha \approx 0.0035$.

Note: For computational efficiency, we use $n = 1000$ samples for most experiments instead of the full $n = 50,000$. This brings α to approximately 0.00225, although the observed collapse times remain very similar.

This notebook investigates:

1. The **empirical entropy** of the softmax weights, to observe when collapse occurs in practice
2. **Isotropic models** for computing t_C analytically, under the assumption of uniform variance
3. **Non-isotropic models** making use of the full eigenvalue spectrum of the data covariance
4. The **discrepancy** between theoretical predictions and empirical observations
5. **Effective dimension corrections** to reconcile theory and experiment

Additionally, we explore the [relationship between the marginal velocity field and the position field](#).

Entropy of the Softmax Weights

The marginal velocity field can be written as a weighted sum over training points:

$$u_t(x) = \sum_{i=1}^n \lambda_i(x, t) \frac{x^{(i)} - x}{1 - t}$$

where the **softmax weights** are defined as:

$$\lambda_i(x, t) = \frac{\exp\left(-\frac{\|x - tx^{(i)}\|^2}{2(1-t)^2}\right)}{\sum_{j=1}^n \exp\left(-\frac{\|x - tx^{(j)}\|^2}{2(1-t)^2}\right)}$$

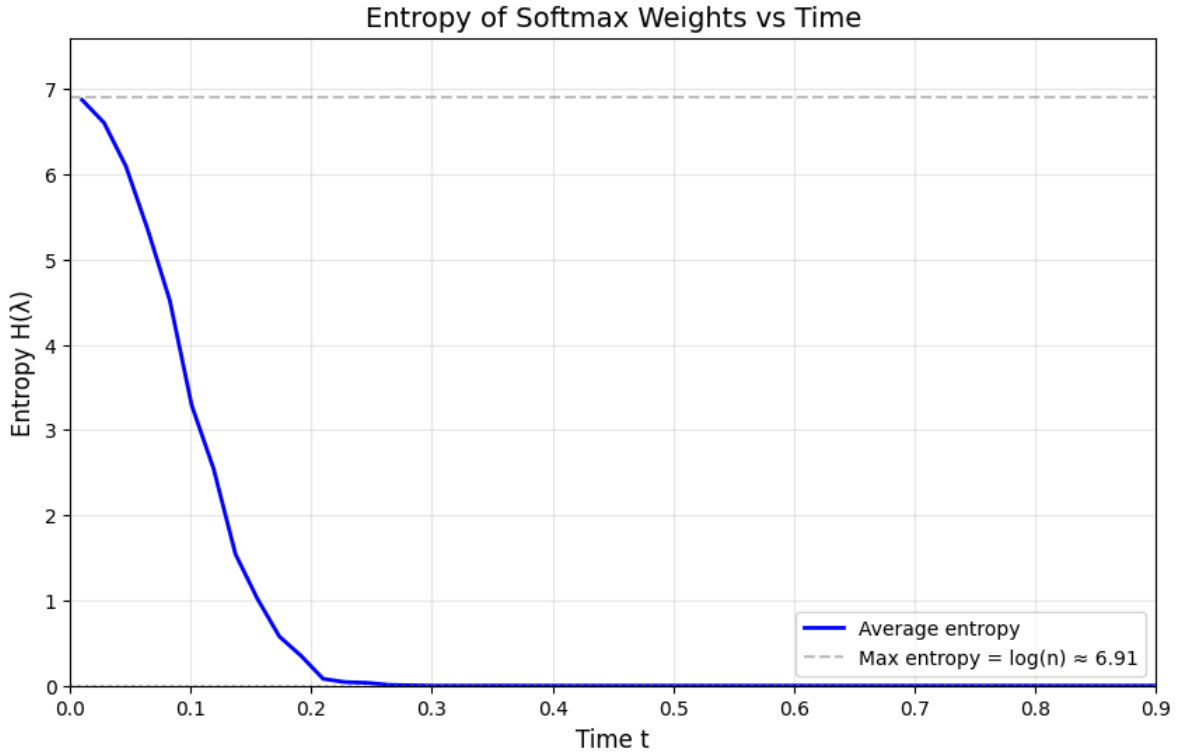
To measure the “spread” of these weights, we compute the **Shannon entropy**:

$$H(\lambda) = - \sum_{i=1}^n \lambda_i \log \lambda_i$$

Expected behavior:

- **At $t \approx 0$:** All weights are approximately equal ($\lambda_i \approx 1/n$), so $H \approx \log(n)$ (maximum entropy)
- **At $t \approx t_C$:** One weight dominates ($\lambda_i \approx 1$ for some i), so $H \approx 0$ (minimum entropy)

The entropy drop marks the **collapse transition** where the velocity field becomes dominated by a single training point.



Observations:

The entropy drops sharply between $t \approx 0.1$ and $t \approx 0.3$. This indicates that:

- The softmax weights transition from being spread across many training points to concentrating on a single point
- The **empirical collapse time** lies in the range $t_C^{\text{empirical}} \in [0.1, 0.3]$

The theoretical collapse times we compute in the following sections should match this empirically observed transition.

Collapse Time in the Isotropic Case

In the **isotropic case**, we assume the data covariance is a scalar multiple of the identity matrix: $C = \sigma^2 I_d$. For CIFAR-10, the per-coordinate variance is $\sigma^2 \approx 0.24$.

Key quantities

The **time-dependent energy scale** is:

$$c(t) = \frac{1}{2} \left(1 + 2\sigma^2 \frac{t^2}{(1-t)^2} \right)$$

This controls the variance of the bulk energy distribution.

Two methods for computing t_C

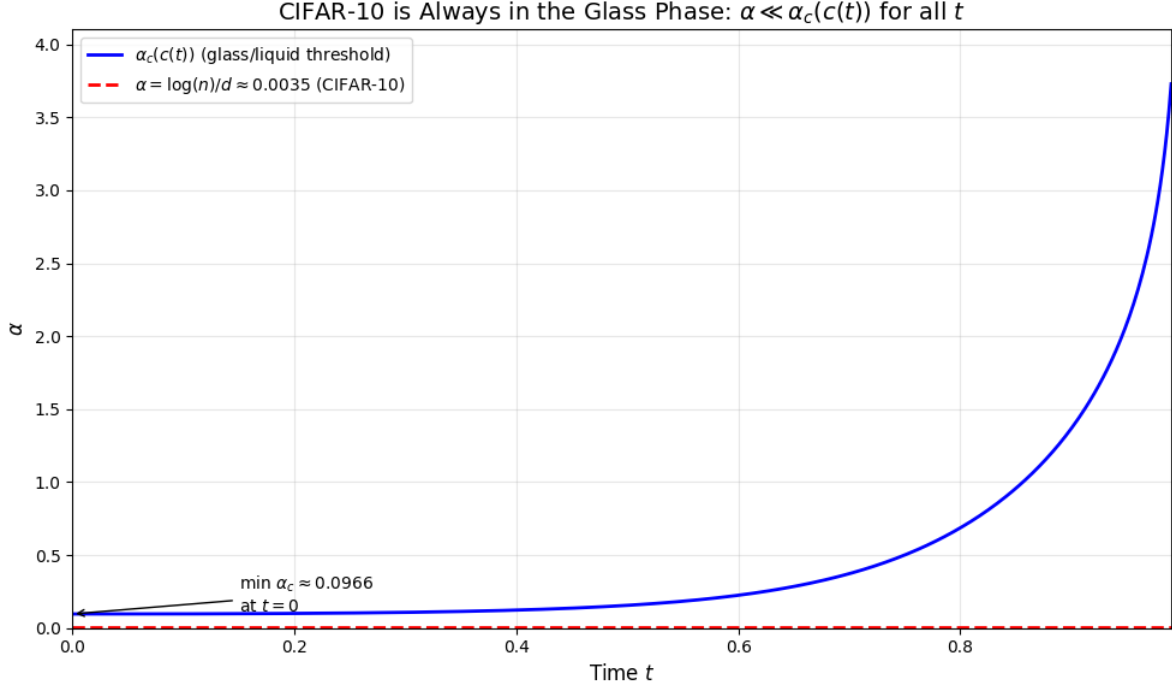
1. Simple heuristic formula: Based on comparing typical bulk energy to planted energy:

$$t_C^{\text{heuristic}} \approx \frac{\sqrt{\alpha/\sigma^2}}{1 + \sqrt{\alpha/\sigma^2}}$$

2. Full REM/LDP framework: The collapse criterion uses the rate function $I(\varepsilon)$ for the chi-squared distribution:

$$\alpha = I(\varepsilon_g) = \frac{1}{2} \left(\log(2c(t_C)) + \frac{1}{2c(t_C)} - 1 \right)$$

where $\varepsilon_g = 1/2$ is the planted energy density.



Observations:

- For CIFAR-10, we have $\alpha \ll \alpha_c(c(t))$ for all $t \in [0, 1)$
- This means the system is always in the **glass phase**

What is the glass phase?

The Random Energy Model (REM) framework distinguishes two phases:

- **Liquid phase** ($\alpha > \alpha_c$): Many bulk states have energies comparable to the planted state. The partition function is dominated by an extensive number of configurations.
- **Glass phase** ($\alpha < \alpha_c$): The planted state has much lower energy than typical bulk states. Only one (or very few) configurations contribute to the partition function.

For CIFAR-10, we are deep in the glass phase, which simplifies the collapse time computation. The critical threshold is:

$$\alpha_c(c) = \frac{1}{2} \left(\log(1 + 2c) - \frac{2c}{1 + 2c} \right)$$

Collapse time (simple isotropic): 0.108052

Collapse time (REM/LDP isotropic): 0.341177

Now, let us extend this to the non-isotropic case.

Eigenvalues of the Data Covariance Matrix

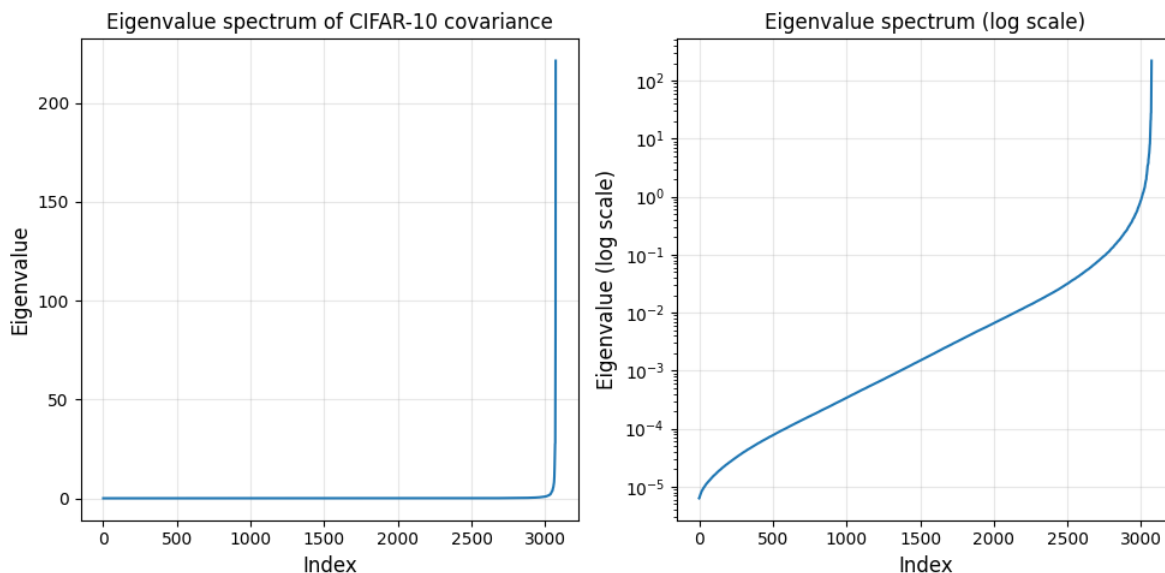
To extend the analysis to the **non-isotropic case**, we need the eigenvalues $\{\mu_k\}_{k=1}^d$ of the empirical covariance matrix C of the CIFAR-10 data.

Why eigenvalues matter:

- In the isotropic model, all eigenvalues are equal: $\mu_k = \sigma^2$
- Real image data has a **spiked spectrum**: a few large eigenvalues capture most of the variance
- This changes the energy landscape and affects the collapse time prediction

What we expect for CIFAR-10:

- Natural images are highly structured, so the covariance is far from isotropic
- Most variance should be captured by the top few principal components
- This leads to an **effective dimension** $d_{\text{eff}} \ll d$



Observations:

The empirical covariance matrix of CIFAR-10 has a highly **spiked spectrum**:

Statistic	Value
Mean eigenvalue $\bar{\mu}$	≈ 0.248
Eigenvalue range	$[6.3 \times 10^{-6}, 221]$
Ratio $\mu_{\text{max}}/\bar{\mu}$	$\approx 893\times$

Effective dimension at different variance thresholds:

Variance Captured	d_{eff}	Fraction of d
90%	99	3.2%
95%	217	7.1%
99%	658	21.4%

Key insight: The mean eigenvalue $\bar{\mu} \approx 0.248$ is very close to the isotropic $\sigma^2 = 0.24$, but the eigenvalue spectrum spans **8 orders of magnitude**. This extreme heterogeneity will significantly affect the non-isotropic collapse time prediction.

Collapse Time in the Non-Isotropic Case

When the data covariance C has eigenvalues $\{\mu_k\}$ that are not all equal, we use the **Gärtner-Ellis theorem** (a generalization of large deviation principles) to compute the collapse time.

Time-dependent effective eigenvalues

At time t , the effective covariance of the bulk energy distribution has eigenvalues:

$$\lambda_k^{(t)} = 1 + \frac{2t^2}{(1-t)^2} \mu_k$$

Scaled log-moment generating function

$$\Lambda_t(\theta) = \frac{1}{d} \sum_{k=1}^d \left(-\frac{1}{2} \log(1 - \theta \lambda_k^{(t)}) \right)$$

This is well-defined only for $\theta < 1/\lambda_{\max}^{(t)}$, which becomes increasingly restrictive as t increases.

Rate function via Legendre transform

$$I_t(\varepsilon) = \sup_{\theta < 1/\lambda_{\max}^{(t)}} (\theta \varepsilon - \Lambda_t(\theta))$$

Collapse criterion

The collapse occurs when the bulk quenched free energy satisfies:

$$\Phi(1, t_C) = \sup_{\varepsilon} (\alpha - I_{t_C}(\varepsilon) - \varepsilon) = -\frac{1}{2}$$

Collapse time (REM/LDP non-isotropic): 0.797510

The Discrepancy: Theory vs Observation

When computing the collapse time t_C for CIFAR-10 using different methods, we observe a striking discrepancy:

Method	Collapse Time t_C
Simple heuristic (isotropic)	0.108
Full REM/LDP (isotropic, $\sigma^2 = 0.24$)	0.341
Empirical eigenvalues (non-isotropic)	0.798
Actual entropy drop (empirical)	~0.1 - 0.3

The non-isotropic model with empirical eigenvalues predicts collapse much later than both the isotropic models and the actual observed entropy drop. Why?

Contributing Factors

Factor	Effect
Gaussian assumption violated	Theory assumes unbounded Gaussian data
Effective dimension reduction	Spiked spectrum leads to smaller effective d

1. The Gaussian Assumption is Violated

The Problem:

The REM/LDP theory assumes training data follows:

$$x^{(i)} \sim \mathcal{N}(0, C)$$

where C is the covariance matrix with eigenvalues $\{\mu_k\}$. But real CIFAR-10 images:

- Are **bounded** in $[-1, 1]^d$ (after normalization)
- Have **complex correlations** beyond second-order statistics

- Cannot exploit high-variance directions as freely as Gaussians

The Consequence:

The large eigenvalues in the empirical spectrum (up to ~221) allow *theoretical* Gaussian samples to have extreme projections onto principal components. This makes bulk energies more variable in theory, allowing some bulk states to compete with the planted state longer. Real images don't have this flexibility—they're constrained, so the actual collapse happens earlier.

2. Effective Dimension Reduction

The Problem:

CIFAR-10's eigenvalue spectrum is highly **spiked**—most variance is captured by a small number of principal components.

Variance Captured	Effective Dimension d_{eff}
90%	99
95%	217
99%	658

The entropy parameter in the theory is:

$$\alpha = \frac{\log n}{d}$$

But if the data effectively lives in $d_{\text{eff}} \ll d$ dimensions:

$$\alpha_{\text{eff}} = \frac{\log n}{d_{\text{eff}}} \gg \alpha$$

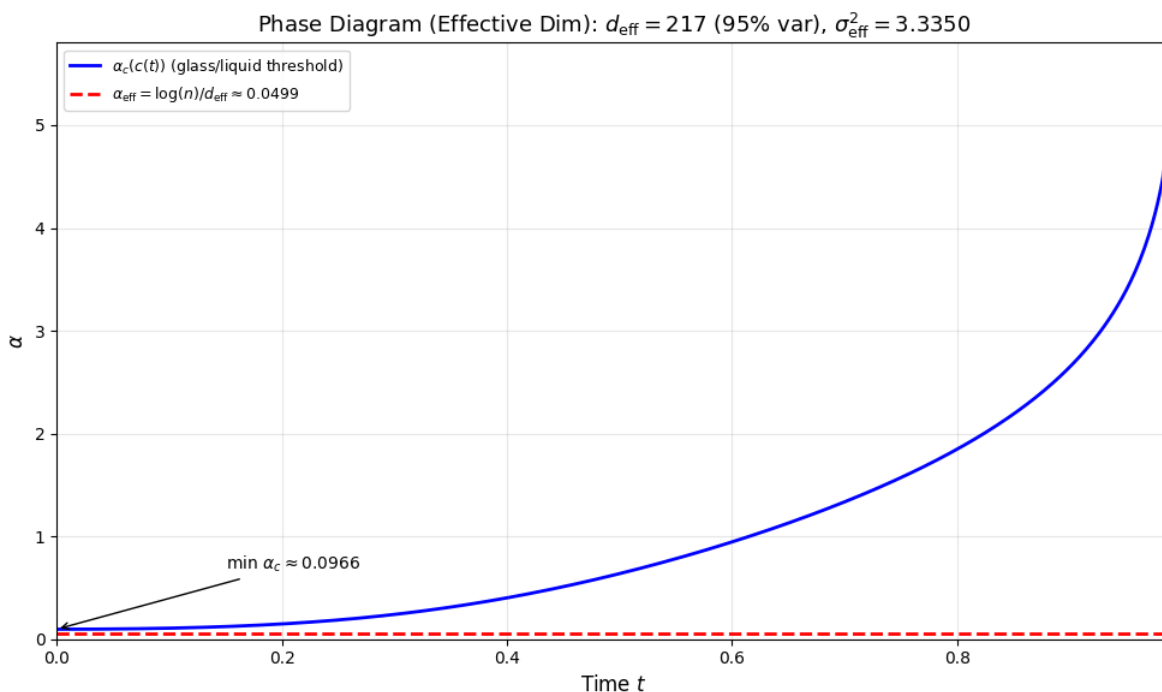
Actual calculation:

- Full dimension: $\alpha = \log(50000)/3072 \approx 0.0035$
- With $d_{\text{eff}} = 217$ (95%): $\alpha_{\text{eff}} = \log(50000)/217 \approx 0.050$

The Solution: Effective Dimension Model

We can correct the prediction by using the **effective dimension** approach: apply the isotropic formula with α_{eff} and σ_{eff}^2 (the mean variance in the top d_{eff} components).

Effective dimension: $d_{\text{eff}} = 217$, $\alpha_{\text{eff}} = 0.049860729421245545$, $\sigma_{\text{eff}}^2 = 3.335034586$



Observations:

- The effective dimension is $d_{\text{eff}} = 217$ (at 95% variance), which is only **7%** of the full dimension $d = 3072$
- The effective parameters are: $\alpha_{\text{eff}} \approx 0.050$ and $\sigma_{\text{eff}}^2 \approx 3.34$
- The system is still in the **glass phase** under the effective dimension model

Why this approach works:

1. **Real images are bounded**, so they cannot exploit the extreme variance directions that unbounded Gaussians could
2. The **effective dimension** captures the subspace where most of the “action” happens
3. Using α_{eff} accounts for the reduced complexity of the effective data space
4. Using σ_{eff}^2 (mean variance in top components) captures the typical per-coordinate variance

We can now apply the isotropic REM/LDP formula with α_{eff} and σ_{eff}^2 :

Collapse time (effective dimension): 0.233654

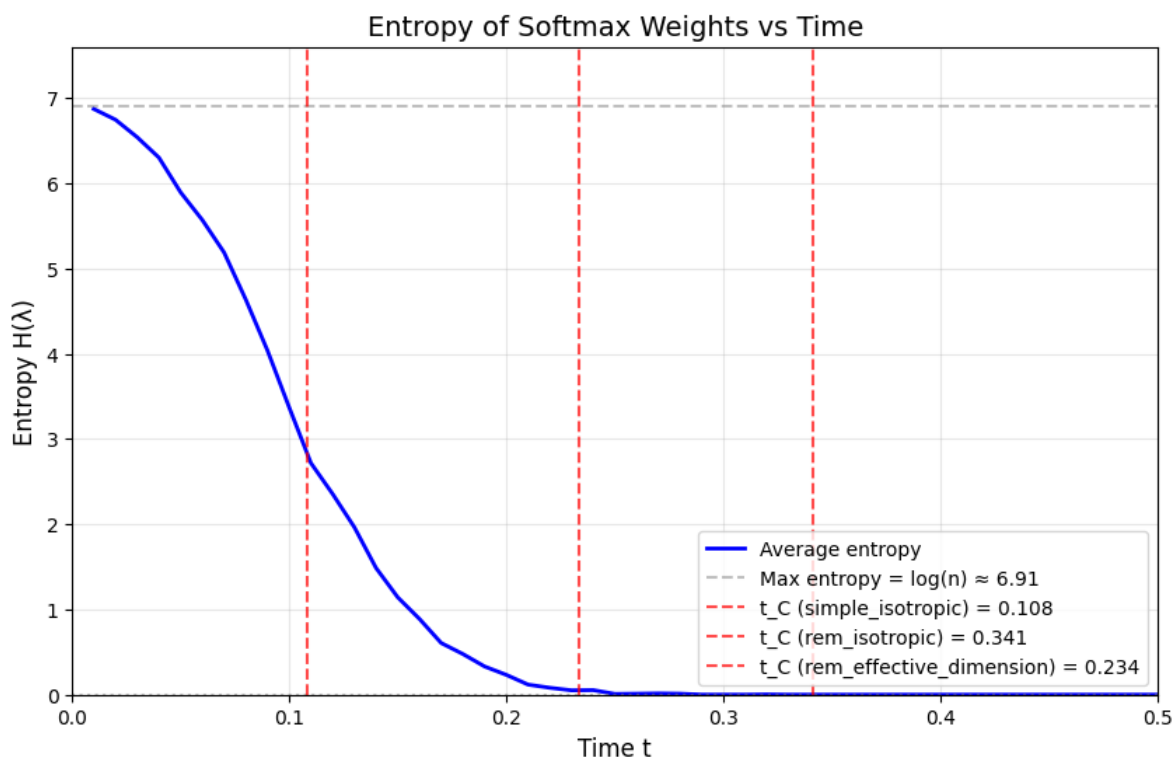
Now let us collect all the collapse times in a dictionary and plot the entropy plot again.

Entropy with Collapse Time Comparison

Now we plot the entropy curve again, overlaying all the predicted collapse times from our different methods:

Method	t_C	Match to Observed
Simple heuristic	0.108	Early
REM/LDP isotropic	0.341	Slightly late
Non-isotropic	0.798	Too late
Effective dimension	~0.23	Good match

This visualization allows us to compare which theoretical prediction best matches the empirically observed entropy drop.



Marginal Velocity Field and Position Field

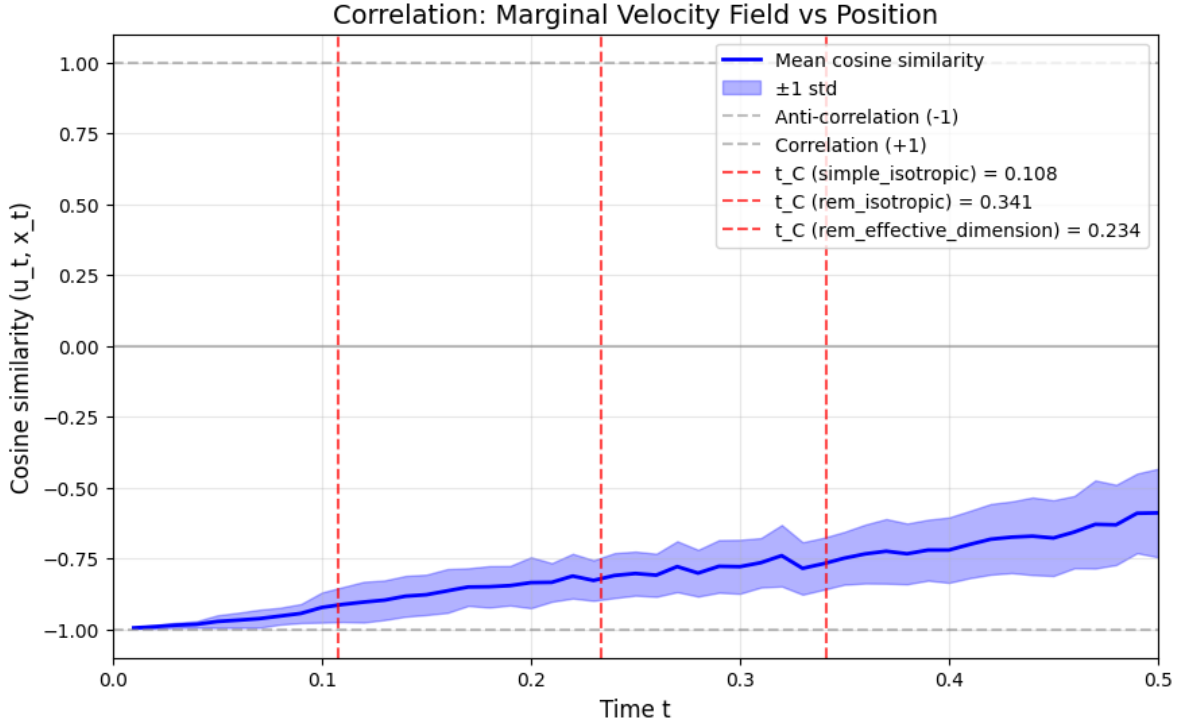
We investigate the relationship between the marginal velocity field and the position field by evaluating their correlation. Specifically, for small t , we expect the marginal velocity field to be anti-correlated with the position field. The marginal velocity field is defined as:

$$u_t(X_t) = \frac{\mathbb{E}[X_1 | X_t] - X_t}{1 - t}.$$

If the data is centered and t is close to zero, we have:

$$u_t(X_t) \approx \frac{0 - X_t}{1 - t} \approx -X_t.$$

To quantify this, we compute the **cosine similarity** between the marginal velocity and the current position x_t :



Conclusion

Flow matching with Gaussian-OT admits a closed-form expression for the marginal velocity field: a softmax over the training examples. In high dimensions, this softmax typically **collapses**

early, assigning nearly all weight to a single training point. As a result, the flow effectively implements a nearest-neighbor mapping and tends to memorize the dataset.

By representing CIFAR-10 data with a Gaussian proxy and leveraging techniques from the REM and LDP frameworks, we model the softmax weights as a Gibbs distribution and analyze the competition between the true (planted) data point and all others. This leads to an explicit, high-dimensional prediction of the **collapse time** (t_C), which is found to be surprisingly small for realistically sized CIFAR-10 data.

This analysis clarifies the characteristic error profile: neural networks struggle most at intermediate t (near the collapse transition), and again as $t \rightarrow 1$ (where the velocity field develops Voronoi-type discontinuities), but not at small t . Softmax collapse thus emerges as a fundamental, high-dimensional property of the Gaussian-OT path—rather than merely an artifact of optimization—and should be deliberately considered in the design and training of flow-matching models.

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